ETH zürich

D INFK



IML Tutorial 2
Linear Regression & Optimization

Anuj Srivastava

27 February 2026

Outline

1. Recap
2. Convexity
3. Closed-form solution for Linear Regression
4. Convergence of Gradient Descent
5. More Losses

Outline

1. Recap

2. Convexity

3. Closed-form solution for Linear Regression

4. Convergence of Gradient Descent

5. More Losses

Recap – Supervised Learning

Training Data → ML method: efficiently find $\hat{f} \in F$ → Model $\hat{f}$

ML Scientists chooses



Function class F
- linear
- nonlinear

Training loss
- squared loss
- asymmetric, Huber

Optimization
- closed-form
- gradient methods

which method (SGD, momentum)

stepsizes

batch sizes

Recap – Linear Regression

Training Data

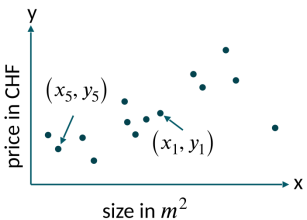
$$\begin{matrix}(x_1, y_1)\\ \vdots \\ (x_n, y_n)\end{matrix}$$

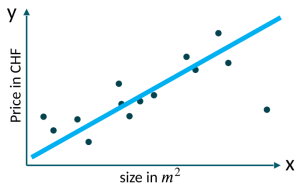
Optimisation problem

$$w^* = \text{argmin}_{w \in \mathbb{R}^2} \frac{1}{2n} \|y - \Phi w\|^2$$

Model $\hat{f}$

$$\hat{f}(x) = x^\top w^*$$





Recap – Linear Regression

1. Define optimization problem (minimizing squared loss): $w^* = \arg\min_w \overbrace{\frac{1}{2}\|y - Xw\|_2^2}^{L(w)}$

Recap – Linear Regression

1. Define optimization problem (minimizing squared loss): $w^* = \arg\min_w \overbrace{\frac{1}{2}\|y - Xw\|_2^2}^{L(w)}$
2. Find the gradient: $\nabla L(w) = X^\top X w - X^\top y$

Recap – Linear Regression

1. Define optimization problem (minimizing squared loss): $w^* = \arg\min_w \overbrace{\frac{1}{2}\|y - Xw\|_2^2}^{L(w)}$
2. Find the gradient: $\nabla L(w) = X^\top X w - X^\top y$

3a. Set gradient to zero $\rightarrow$ **Closed-form solution**: $\nabla L(w^*) \stackrel{!}{=} 0 \implies w^* = (X^\top X)^{-1} X^\top y$

Recap – Linear Regression

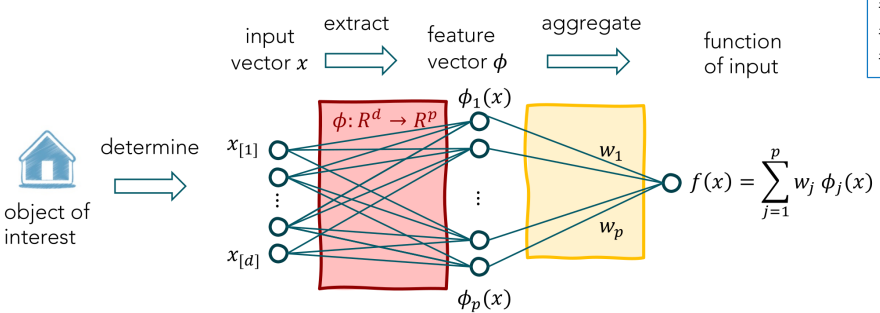
1. Define optimization problem (minimizing squared loss): $w^* = \arg\min_w \overbrace{\frac{1}{2}\|y - Xw\|_2^2}^{L(w)}$
2. Find the gradient: $\nabla L(w) = X^\top X w - X^\top y$

3a. Set gradient to zero $\rightarrow$ **Closed-form solution**: $\nabla L(w^*) \stackrel{!}{=} 0 \implies w^* = (X^\top X)^{-1} X^\top y$

3b. Perform **Gradient Descent**: At step t, $w^{t+1} = w^t - \eta \nabla L(w^t)$

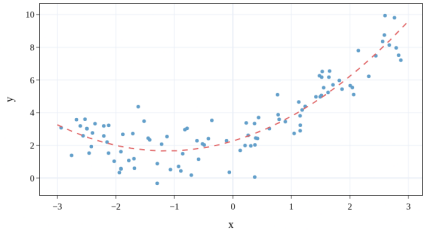
Recap – Non-Linear Regression

Many types of nonlinear functions of input vector x can be written as



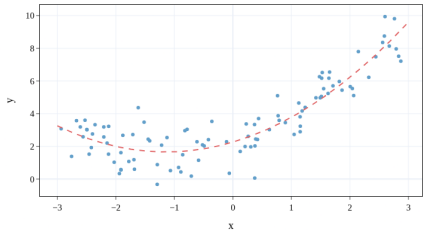
- Some ML models fit data with fixed mappings ϕ: Linear regression, polynomial and kernel regression
- Others also learn the mappings ϕ: when fitting the data: neural networks

Non-Linear Example – Polynomial functions



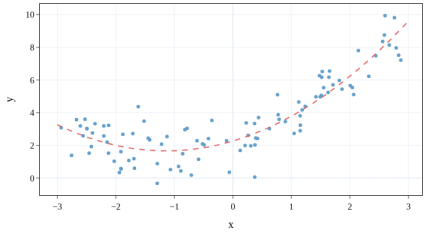
Non-Linear Example – Polynomial functions

1. Define $\phi(x) = [1, x, x^2]^\top$



Non-Linear Example – Polynomial functions

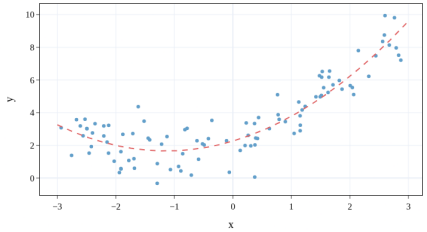
1. Define $\phi(x) = [1, x, x^2]^\top$
2. We can then write $\hat{f}(x) = w^\top \phi(x)$



Non-Linear Example – Polynomial functions

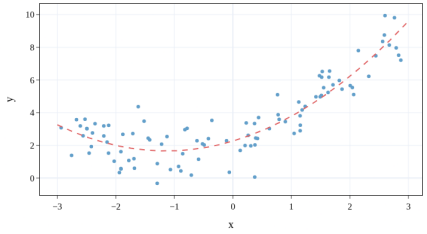
1. Define $\phi(x) = [1, x, x^2]^\top$
2. We can then write $\hat{f}(x) = w^\top \phi(x)$
3. The design matrix becomes

$$\Phi = \begin{bmatrix} 1 & x_1 & x_1^2 \\ \vdots & & \vdots \\ 1 & x_n & x_n^2 \end{bmatrix}$$

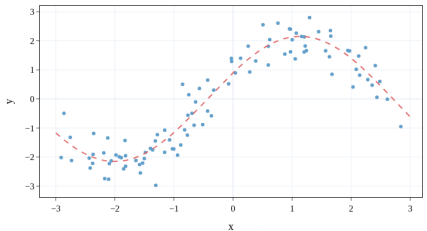


Non-Linear Example – Polynomial functions

1. Define $\phi(x) = [1, x, x^2]^\top$
2. We can then write $\hat{f}(x) = w^\top \phi(x)$
3. The design matrix becomes
$$\Phi = \begin{bmatrix} 1 & x_1 & x_1^2 \\ \vdots & & \vdots \\ 1 & x_n & x_n^2 \end{bmatrix}$$
4. Set gradient to zero:
$$\nabla L(w^*) \stackrel{!}{=} 0$$
$$\implies w^* = (\Phi^\top \Phi)^{-1} \Phi^\top y$$

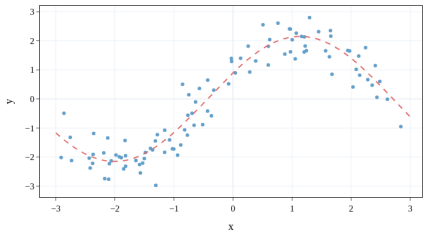


Non-Linear Example – Trigonometric functions



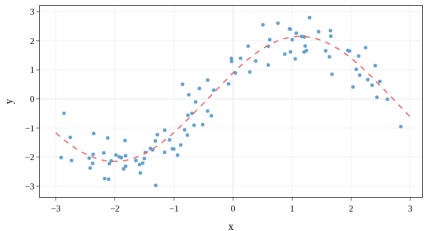
Non-Linear Example – Trigonometric functions

1. Define $\phi(x) = [\sin(x), \cos(x)]^\top$



Non-Linear Example – Trigonometric functions

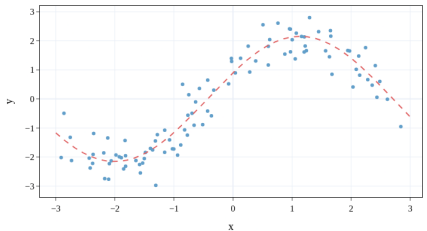
1. Define $\phi(x) = [\sin(x), \cos(x)]^\top$
2. We can then write $\hat{f}(x) = w^\top \phi(x)$



Non-Linear Example – Trigonometric functions

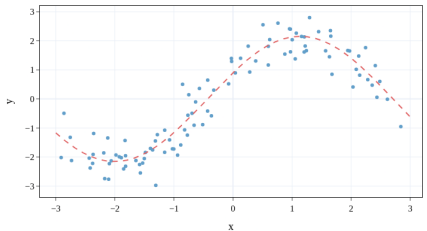
1. Define $\phi(x) = [\sin(x), \cos(x)]^\top$
2. We can then write $\hat{f}(x) = w^\top \phi(x)$
3. The design matrix becomes

$$\Phi = \begin{bmatrix} \sin(x_1) & \cos(x_1) \\ \vdots & \vdots \\ \sin(x_n) & \cos(x_n) \end{bmatrix}$$



Non-Linear Example – Trigonometric functions

1. Define $\phi(x) = [\sin(x), \cos(x)]^\top$
2. We can then write $\hat{f}(x) = w^\top \phi(x)$
3. The design matrix becomes
$$\Phi = \begin{bmatrix} \sin(x_1) & \cos(x_1) \\ \vdots & \vdots \\ \sin(x_n) & \cos(x_n) \end{bmatrix}$$
4. Set gradient to zero:
$$\nabla L(w^*) \stackrel{!}{=} 0$$
$$\implies w^* = (\Phi^\top \Phi)^{-1} \Phi^\top y$$



See `non-linear-bases.ipynb`

Recap – Gradient Descent

Gradient descent algorithm to minimize $L(w)$

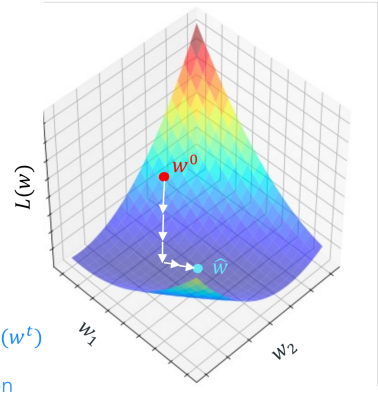
Start at initial w^0

for simplicity, fixed for all t

At each step t, $w^{t+1} = w^t - \eta \nabla_w L(w^t), t \leftarrow t+1$

If e.g. $|L(w^t) - L(w^{t-1})| > \epsilon$, continue

Else stop and output $\hat{w} = w^t$



Linear approximation for the loss value at step $t+1$

$$L(w^{t+1}) = L(w^t - \eta \nabla L(w^t)) \approx L(w^t) - \eta \langle \nabla L(w^t), \nabla L(w^t) \rangle < L(w^t)$$

the negative gradient direction is a descent direction

for small enough stepsize η (see lecture notes Theorem 5.2.)

Gradient Descent – Taylor expansion perspective

1. Multivariate Taylor expansion reads $L(w^{t+1}) = L(w^t) - \eta ||\nabla L(w^t)||_2^2 + \xi(-\eta \nabla L(w^t))$
2. By definition of (directional) derivative for any vector v we have $\lim_{\eta \to 0} \frac{L(w+\eta v) - L(w)}{\eta} - \langle \nabla L(w), v \rangle = 0$
3. Hence plugging in $w = w^t$, $\mathrm{v} = -\nabla L(w^t)$ and remembering $w^{t+1} = w^t + \eta v$ we see that the

   remainder term satisfies $\lim_{\eta \to 0} \frac{\xi\left(-\eta \, \nabla L(w^t)\right)}{\eta} = 0$ and by definition of convergence,

   for any ϵ, there exists η small enough, s.t. $|\xi\left(-\eta \, \nabla L(w^t)\right)| \leq \epsilon \, \eta$
4. Now pick $\epsilon < \left|\left|\nabla L(w^t)\right|\right|_2^2$. Then by the step above, there exists η that yields

   $$L(w^{t+1}) \leq L(w^t) - \eta \left|\left|\nabla L(w^t)\right|\right|_2^2 + \epsilon \, \eta < L(w^t)$$

   the negative gradient direction is a descent direction (for small enough stepsize)

Outline

1. Recap

2. Convexity

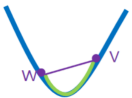
3. Closed-form solution for Linear Regression

4. Convergence of Gradient Descent

5. More Losses

Convex functions

We can characterize convex functions in different ways:

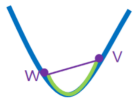


0-th order condition: function lies below chords

$$L(\lambda w + (1-\lambda)v) \leq \lambda L(w) + (1-\lambda)L(v)$$

Convex functions

We can characterize convex functions in different ways:



0-th order condition: function lies below chords

$$L(\lambda w + (1-\lambda)v) \leq \lambda L(w) + (1-\lambda)L(v)$$

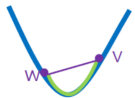


1st order condition: function lies above tangents

$$L(v) \geq L(w) + \nabla L(w)^\top (v - w)$$

Convex functions

We can characterize convex functions in different ways:



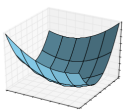
0-th order condition: function lies below chords

$$L(\lambda w + (1-\lambda)v) \leq \lambda L(w) + (1-\lambda)L(v)$$



1st order condition: function lies above tangents

$$L(v) \geq L(w) + \nabla L(w)^\top (v - w)$$



2nd order condition: non-negative curvature (positive semi-definite Hessian)

$$\nabla^2 L(w) \succeq 0$$

Strongly Convex functions

m-**strongly convex** functions (for $m > 0$)

- 0-th and 1st order conditions have additional quadratic terms (only 1st order shown):

$$L(v) \geq L(w) + \nabla L(w)^\top (v - w) + \frac{m}{2}\|w - v\|_2^2$$

Strongly Convex functions

m-**strongly convex** functions (for $m > 0$)

- 0-th and 1st order conditions have additional quadratic terms (only 1st order shown):

$$L(v) \geq L(w) + \nabla L(w)^\top (v - w) + \frac{m}{2}\|w - v\|_2^2$$

- 2nd order condition: $\nabla^2 L(w) \succeq mI$

Strongly Convex functions

m-**strongly convex** functions (for $m > 0$)

- 0-th and 1st order conditions have additional quadratic terms (only 1st order shown):

$$L(v) \geq L(w) + \nabla L(w)^\top (v - w) + \frac{m}{2}\|w - v\|_2^2$$

- 2nd order condition: $\nabla^2 L(w) \succeq mI$
- Global minimum is **unique**

Strongly Convex functions

m-**strongly convex** functions (for $m > 0$)

- 0-th and 1st order conditions have additional quadratic terms (only 1st order shown):

$$L(v) \geq L(w) + \nabla L(w)^\top (v - w) + \frac{m}{2}\|w - v\|_2^2$$

- 2nd order condition: $\nabla^2 L(w) \succeq mI$
- Global minimum is **unique**
- (Bonus) gradient descent **converges fast**

Minimizing Convex functions

Stationary points $(\nabla L(w^*) = 0)$ minimize convex functions – why?

Minimizing Convex functions

Stationary points $(\nabla L(w^*) = 0)$ minimize convex functions – why?

1st order condition at w^*: for any v,

$$L(v) \geq L(w^*) + \underbrace{\nabla L(w^*)^\top}_{=0}(v - w^*) = L(w^*)$$

Minimizing Convex functions

Stationary points $(\nabla L(w^*) = 0)$ minimize convex functions – why?

1st order condition at w^*: for any v,

$$L(v) \geq L(w^*) + \underbrace{\nabla L(w^*)^\top}_{=0}(v - w^*) = L(w^*)$$

Under strong convexity w^* is **unique** – since for any $v \neq w^*$,

$$\begin{aligned} L(v) &\geq L(w^*) + \underbrace{\nabla L(w^*)^\top}_{=0}(v - w^*) + \underbrace{\frac{m}{2}\|v - w^*\|_2^2}_{>0} \\ &> L(w^*) \end{aligned}$$

Outline

1. Recap

2. Convexity

3. Closed-form solution for Linear Regression

4. Convergence of Gradient Descent

5. More Losses

Closed-form solution for Linear Regression

Recall:

- Squared loss for linear regression is convex: $L(w) = \frac{1}{2}\|y - Xw\|_2^2$
  - because $\|\cdot\|_2^2$ is convex and $(y - Xw)$ is affine
- Hence, stationary points are minimizers: $0 \overset{!}{=} \nabla L(w^*) = X^\top X w^* - X^\top y$
  - Also, note that $\nabla^2 L(w) = X^\top X$
- If $X^\top X$ is invertible, then we have the closed-form $w^* = (X^\top X)^{-1} X^\top y$

Closed-form solution for Linear Regression

Recall:

- Squared loss for linear regression is convex: $L(w) = \frac{1}{2}\|y - Xw\|_2^2$
  - because $\|\cdot\|_2^2$ is convex and $(y - Xw)$ is affine
- Hence, stationary points are minimizers: $0 \overset{!}{=} \nabla L(w^*) = X^\top X w^* - X^\top y$
  - Also, note that $\nabla^2 L(w) = X^\top X$
- If $X^\top X$ is invertible, then we have the closed-form $w^* = (X^\top X)^{-1} X^\top y$

When is $X^\top X$ is invertible? Is $L(w)$ strongly convex?

Closed-form solution for Linear Regression

So far, we have assumed $d \leq n$ (underparameterized)

Closed-form solution for Linear Regression

So far, we have assumed $d \leq n$ (underparameterized)

- If the SVD of $X = U\Sigma V^\top$, then SVD of $X^\top X = V\Sigma^2 V^\top$

Closed-form solution for Linear Regression

So far, we have assumed $d \leq n$ (underparameterized)

- If the SVD of $X = U\Sigma V^\top$, then SVD of $X^\top X = V\Sigma^2 V^\top$
- And if invertible, $(X^\top X)^{-1} = V\Sigma^{-2}V^\top$

Closed-form solution for Linear Regression

So far, we have assumed $d \leq n$ (underparameterized)

- If the SVD of $X = U\Sigma V^\top$, then SVD of $X^\top X = V\Sigma^2 V^\top$
- And if invertible, $(X^\top X)^{-1} = V\Sigma^{-2}V^\top \leftarrow$ exists if and only if each $\sigma_i \neq 0$, i.e. X is **full-rank**

Closed-form solution for Linear Regression

So far, we have assumed $d \leq n$ (underparameterized)

- If the SVD of $X = U\Sigma V^\top$, then SVD of $X^\top X = V\Sigma^2 V^\top$
- And if invertible, $(X^\top X)^{-1} = V\Sigma^{-2}V^\top \leftarrow$ exists if and only if each $\sigma_i \neq 0$, i.e. X is **full-rank**
  - If each $\sigma_i \neq 0$, then $\lambda_{\min}(X^\top X) > 0$

Closed-form solution for Linear Regression

So far, we have assumed $d \leq n$ (underparameterized)

- If the SVD of $X = U\Sigma V^\top$, then SVD of $X^\top X = V\Sigma^2 V^\top$
- And if invertible, $(X^\top X)^{-1} = V\Sigma^{-2}V^\top \leftarrow$ exists if and only if each $\sigma_i \neq 0$, i.e. X is **full-rank**
  - If each $\sigma_i \neq 0$, then $\lambda_{\min}(X^\top X) > 0$
  - $\nabla^2 L(w) = X^\top X \succeq \lambda_{\min} I \implies L(w)$ is $\lambda_{\min}$-strongly convex!

Closed-form solution for Linear Regression

So far, we have assumed $d \leq n$ (underparameterized)

- If the SVD of $X = U\Sigma V^\top$, then SVD of $X^\top X = V\Sigma^2 V^\top$
- And if invertible, $(X^\top X)^{-1} = V\Sigma^{-2}V^\top \leftarrow$ exists if and only if each $\sigma_i \neq 0$, i.e. X is **full-rank**
  - If each $\sigma_i \neq 0$, then $\lambda_{\min}(X^\top X) > 0$
  - $\nabla^2 L(w) = X^\top X \succeq \lambda_{\min} I \implies L(w)$ is $\lambda_{\min}$-strongly convex!

What if $n < d$ (overparameterized), or more generally, X is **not** full-rank?

Overparameterized or Rank-deficient Linear Regression

If $n < d$ (overparameterization) or if X is rank-deficient in general:

- We have $\operatorname{rank}(X^\top X) = \operatorname{rank}(X) < d$, so $X^\top X \in \mathbb{R}^{d \times d}$ is not invertible

Overparameterized or Rank-deficient Linear Regression

If $n < d$ (overparameterization) or if X is rank-deficient in general:

- We have $\mathrm{rank}(X^\top X) = \mathrm{rank}(X) < d$, so $X^\top X \in \mathbb{R}^{d \times d}$ is not invertible
- Further, X has a non-trivial null space / kernel:

$$\ker(X) = \{u \in \mathbb{R}^d \mid Xu = 0\}$$

Overparameterized or Rank-deficient Linear Regression

If $n < d$ (overparameterization) or if X is rank-deficient in general:

- We have $\operatorname{rank}(X^\top X) = \operatorname{rank}(X) < d$, so $X^\top X \in \mathbb{R}^{d \times d}$ is not invertible
- Further, X has a non-trivial null space / kernel:

$$\ker(X) = \{u \in \mathbb{R}^d \mid Xu = 0\}$$

- Recall: for optimal solution w^*, $X^\top X w^* = X^\top y$. But for any $u \in \ker(X)$:

$$X^\top X(w^* + u) = X^\top X w^* = X^\top y$$

Overparameterized or Rank-deficient Linear Regression

If $n < d$ (overparameterization) or if X is rank-deficient in general:

- We have $\operatorname{rank}(X^\top X) = \operatorname{rank}(X) < d$, so $X^\top X \in \mathbb{R}^{d \times d}$ is not invertible
- Further, X has a non-trivial null space / kernel:

$$\ker(X) = \{u \in \mathbb{R}^d \mid Xu = 0\}$$

- Recall: for optimal solution w^*, $X^\top X w^* = X^\top y$. But for any $u \in \ker(X)$:

$$X^\top X(w^* + u) = X^\top X w^* = X^\top y$$

- So any $w^* + u$ is also optimal. Infinite solutions!

Overparameterized or Rank-deficient Linear Regression

If $n < d$ (overparameterization) or if X is rank-deficient in general:

- We have $\text{rank}(X^\top X) = \text{rank}(X) < d$, so $X^\top X \in \mathbb{R}^{d \times d}$ is not invertible
- Further, X has a non-trivial null space / kernel:

$$\ker(X) = \{u \in \mathbb{R}^d \mid Xu = 0\}$$

- Recall: for optimal solution w^*, $X^\top X w^* = X^\top y$. But for any $u \in \ker(X)$:

$$X^\top X(w^* + u) = X^\top X w^* = X^\top y$$

- So any $w^* + u$ is also optimal. Infinite solutions!

How to pick a unique solution?

Overparameterized or Rank-deficient Linear Regression

If $n < d$ (overparameterization) or if X is rank-deficient in general:

- We have $\text{rank}(X^\top X) = \text{rank}(X) < d$, so $X^\top X \in \mathbb{R}^{d \times d}$ is not invertible
- Further, X has a non-trivial null space / kernel:

$$\ker(X) = \{u \in \mathbb{R}^d \mid Xu = 0\}$$

- Recall: for optimal solution w^*, $X^\top X w^* = X^\top y$. But for any $u \in \ker(X)$:

$$X^\top X(w^* + u) = X^\top X w^* = X^\top y$$

- So any $w^* + u$ is also optimal. Infinite solutions!

How to pick a unique solution? Find one with the **smallest norm**:

$$\hat{w} = \arg\min_w \|w\|_2 \quad \text{s.t. } X^\top X w = X^\top y$$

Moore-Penrose Inverse / Pseudo-inverse

For any matrix $A \in \mathbb{R}^{m \times n}$, its pseudo-inverse is a matrix $A^\dagger \in \mathbb{R}^{n \times m}$ that satisfies:

- $AA^\dagger A = A$ (that is, $AA^\dagger$ acts like identity on $\mathrm{span}(A)$)
- $A^\dagger AA^\dagger = A^\dagger$
- $AA^\dagger$ and $A^\dagger A$ are symmetric

Moore-Penrose Inverse / Pseudo-inverse

For any matrix $A \in \mathbb{R}^{m \times n}$, its pseudo-inverse is a matrix $A^\dagger \in \mathbb{R}^{n \times m}$ that satisfies:

- $AA^\dagger A = A$ (that is, $AA^\dagger$ acts like identity on $\mathrm{span}(A)$)
- $A^\dagger AA^\dagger = A^\dagger$
- $AA^\dagger$ and $A^\dagger A$ are symmetric

The pseudo-inverse always exists, is unique, and further has the following properties:

- $(A^\top)^\dagger = (A^\dagger)^\top; (AA^\top)^\dagger = (A^\dagger)^\top A^\dagger$ (similar to the inverse)

Moore-Penrose Inverse / Pseudo-inverse

For any matrix $A \in \mathbb{R}^{m \times n}$, its pseudo-inverse is a matrix $A^\dagger \in \mathbb{R}^{n \times m}$ that satisfies:

- $AA^\dagger A = A$ (that is, $AA^\dagger$ acts like identity on $\mathrm{span}(A)$)
- $A^\dagger AA^\dagger = A^\dagger$
- $AA^\dagger$ and $A^\dagger A$ are symmetric

The pseudo-inverse always exists, is unique, and further has the following properties:

- $(A^\top)^\dagger = (A^\dagger)^\top; (AA^\top)^\dagger = (A^\dagger)^\top A^\dagger$ (similar to the inverse)
- has same null-space as $A^\top$: $A^\top x = 0 \iff A^\dagger x = 0$

Moore-Penrose Inverse / Pseudo-inverse

For any matrix $A \in \mathbb{R}^{m \times n}$, its pseudo-inverse is a matrix $A^\dagger \in \mathbb{R}^{n \times m}$ that satisfies:

- $AA^\dagger A = A$ (that is, $AA^\dagger$ acts like identity on $\mathrm{span}(A)$)
- $A^\dagger AA^\dagger = A^\dagger$
- $AA^\dagger$ and $A^\dagger A$ are symmetric

The pseudo-inverse always exists, is unique, and further has the following properties:

- $(A^\top)^\dagger = (A^\dagger)^\top; (AA^\top)^\dagger = (A^\dagger)^\top A^\dagger$ (similar to the inverse)
- has same null-space as $A^\top$: $A^\top x = 0 \iff A^\dagger x = 0$
- Easy to express with SVD: $A = U\Sigma V^\top \implies A^\dagger = V\Sigma^\dagger U^\top$, where:

$$\Sigma^\dagger_{ii} = \begin{cases} \sigma_i^{-1}, & \text{if } \sigma_i \neq 0, \\ 0, & \text{else} \end{cases}$$

Min-norm Solution to Linear Regression

The min-norm solution is obtained using the pseudo-inverse: $\hat{w} = (X^\top X)^\dagger X^\top y = X^\dagger y$

- Proof on blackboard (and lecture notes, Section 4.2.3)

Min-norm Solution to Linear Regression

The min-norm solution is obtained using the pseudo-inverse: $\hat{w} = (X^\top X)^\dagger X^\top y = X^\dagger y$

- Proof on blackboard (and lecture notes, Section 4.2.3)
- If we initialize $w^0 = 0$, gradient descent converges to $\hat{w}$! (upcoming in Homework 1 - Exercise 2)

Min-norm Solution to Linear Regression

The min-norm solution is obtained using the pseudo-inverse: $\hat{w} = (X^\top X)^\dagger X^\top y = X^\dagger y$

- Proof on blackboard (and lecture notes, Section 4.2.3)
- If we initialize $w^0 = 0$, gradient descent converges to $\hat{w}$! (upcoming in Homework 1 - Exercise 2)

If X is full-rank but overparameterized ($n < d$, $\mathrm{rank}(X) = n$), then:

- $XX^\top \in \mathbb{R}^{n \times n}$ is invertible !

Min-norm Solution to Linear Regression

The min-norm solution is obtained using the pseudo-inverse: $\hat{w} = (X^\top X)^\dagger X^\top y = X^\dagger y$

- Proof on blackboard (and lecture notes, Section 4.2.3)
- If we initialize $w^0 = 0$, gradient descent converges to $\hat{w}$! (upcoming in Homework 1 - Exercise 2)

If X is full-rank but overparameterized ($n < d$, $\mathrm{rank}(X) = n$), then:

- $XX^\top \in \mathbb{R}^{n \times n}$ is invertible !
- By properties of pseudo-inverse, $X^\dagger = X^\top (XX^\top)^{-1}$, thus $\hat{w} = X^\top (XX^\top)^{-1} y$

Outline

1. Recap

2. Convexity

3. Closed-form solution for Linear Regression

4. Convergence of Gradient Descent

5. More Losses

Convergence of Gradient Descent for Linear Regression

- Recall gradient descent: $w^{t+1} = w^t - \eta \nabla L(w^t)$

Convergence of Gradient Descent for Linear Regression

- Recall gradient descent: $w^{t+1} = w^t - \eta \nabla L(w^t)$
- For linear regression: $\nabla L(w) = X^\top X w - X^\top y$, hence:

$$w^{t+1} = w^t - \eta X^\top X w^t + \eta X^\top y$$

Convergence of Gradient Descent for Linear Regression

- Recall gradient descent: $w^{t+1} = w^t - \eta \nabla L(w^t)$
- For linear regression: $\nabla L(w) = X^\top X w - X^\top y$, hence:

$$w^{t+1} = w^t - \eta X^\top X w^t + \eta X^\top y$$

- Can be rewritten as (shown on blackboard):

$$w^{t+1} - w^* = (I - \eta X^\top X)(w^t - w^*)$$

Convergence of Gradient Descent for Linear Regression

- Recall gradient descent: $w^{t+1} = w^t - \eta \nabla L(w^t)$
- For linear regression: $\nabla L(w) = X^\top X w - X^\top y$, hence:

$$w^{t+1} = w^t - \eta X^\top X w^t + \eta X^\top y$$

- Can be rewritten as (shown on blackboard):

$$w^{t+1} - w^* = (I - \eta X^\top X)(w^t - w^*)$$
$$\implies \|w^{t+1} - w^*\|_2 \leq \underbrace{\left\|I - \eta X^\top X\right\|_{\text{op}}}_{=\rho} \|w^t - w^*\|_2$$

Convergence of Gradient Descent for Linear Regression

- Recall gradient descent: $w^{t+1} = w^t - \eta \nabla L(w^t)$
- For linear regression: $\nabla L(w) = X^\top X w - X^\top y$, hence:

$$w^{t+1} = w^t - \eta X^\top X w^t + \eta X^\top y$$

- Can be rewritten as (shown on blackboard):

$$\begin{aligned} w^{t+1} - w^* &= (I - \eta X^\top X)(w^t - w^*) \\ \implies \|w^{t+1} - w^*\|_2 &\leq \underbrace{\left\|I - \eta X^\top X\right\|_{\text{op}}}_{=\rho} \|w^t - w^*\|_2 \\ &\vdots \\ &\leq \rho^{t+1} \|w^0 - w^*\|_2 \end{aligned}$$

Convergence of Gradient Descent for Linear Regression

- Recall gradient descent: $w^{t+1} = w^t - \eta \nabla L(w^t)$
- For linear regression: $\nabla L(w) = X^\top X w - X^\top y$, hence:

$$w^{t+1} = w^t - \eta X^\top X w^t + \eta X^\top y$$

- Can be rewritten as (shown on blackboard):

$$\begin{aligned} w^{t+1} - w^* &= (I - \eta X^\top X)(w^t - w^*) \\ \implies \|w^{t+1} - w^*\|_2 &\leq \underbrace{\left\|I - \eta X^\top X\right\|_{\text{op}}}_{=\rho} \|w^t - w^*\|_2 \\ &\vdots \\ &\leq \rho^{t+1} \|w^0 - w^*\|_2 \end{aligned}$$

- Distance of estimate w^t from optimum w^* is exponential ($C\rho^t$) ! **When** is $\rho < 1$?

Convergence of Gradient Descent for Linear Regression

When is $\rho = \|I - \eta X^\top X\|_{\mathrm{op}} < 1$?

Convergence of Gradient Descent for Linear Regression

When is $\rho = \|I - \eta X^\top X\|_{\text{op}} < 1$?

- $\rho = \max\left\{\left|1 - \eta\lambda_{\min}(X^\top X)\right|, \left|1 - \eta\lambda_{\max}(X^\top X)\right|\right\}$ (on blackboard, visualization)

Convergence of Gradient Descent for Linear Regression

When is $\rho = \|I - \eta X^\top X\|_{\text{op}} < 1$?

- $\rho = \max\left\{\left|1 - \eta\lambda_{\min}(X^\top X)\right|, \left|1 - \eta\lambda_{\max}(X^\top X)\right|\right\}$ (on blackboard, visualization)
- $\rho < 1$ only if $0 < \eta < 2/\lambda_{\max}$
- ρ is minimized when $\eta^* = 2/(\lambda_{\max} + \lambda_{\min})$, and

$$\rho^* = \frac{\lambda_{\max} - \lambda_{\min}}{\lambda_{\max} + \lambda_{\min}} = \frac{\kappa - 1}{\kappa + 1}$$

  ρ^* only depends on the ratio $\kappa = \lambda_{\max}/\lambda_{\min}$!

Convergence of Gradient Descent for Linear Regression

When is $\rho = \|I - \eta X^\top X\|_{\text{op}} < 1$?

- $\rho = \max\left\{\left|1 - \eta\lambda_{\min}(X^\top X)\right|, \left|1 - \eta\lambda_{\max}(X^\top X)\right|\right\}$ (on blackboard, visualization)
- $\rho < 1$ only if $0 < \eta < 2/\lambda_{\max}$
- ρ is minimized when $\eta^* = 2/(\lambda_{\max} + \lambda_{\min})$, and

$$\rho^* = \frac{\lambda_{\max} - \lambda_{\min}}{\lambda_{\max} + \lambda_{\min}} = \frac{\kappa - 1}{\kappa + 1}$$

  ρ^* only depends on the ratio $\kappa = \lambda_{\max}/\lambda_{\min}$!

- Large ratio: $\kappa \to \infty \implies \rho^* \to 1$ (slow convergence, see `gradient-descent.ipynb`)

Convergence of Gradient Descent for Linear Regression

When is $\rho = \|I - \eta X^\top X\|_{\text{op}} < 1$?

- $\rho = \max\left\{\left|1 - \eta\lambda_{\min}(X^\top X)\right|, \left|1 - \eta\lambda_{\max}(X^\top X)\right|\right\}$ (on blackboard, visualization)
- $\rho < 1$ only if $0 < \eta < 2/\lambda_{\max}$
- ρ is minimized when $\eta^* = 2/(\lambda_{\max} + \lambda_{\min})$, and

$$\rho^* = \frac{\lambda_{\max} - \lambda_{\min}}{\lambda_{\max} + \lambda_{\min}} = \frac{\kappa - 1}{\kappa + 1}$$

  ρ^* only depends on the ratio $\kappa = \lambda_{\max}/\lambda_{\min}$!

- Large ratio: $\kappa \to \infty \implies \rho^* \to 1$ (slow convergence, see `gradient-descent.ipynb`)
- Identical eigenvalues: $\kappa = 1 \implies \rho^* = 0$ (optimum reached in one step!)

(Bonus) Convergence of Gradient Descent for Strongly Convex Functions

(Bonus) Convergence of Gradient Descent for Strongly Convex Functions

- To establish convergence of m-strongly convex functions:

$$L(v) \geq L(w) + \nabla L(w)^\top (v - w) + \frac{m}{2}\|v - w\|_2^2$$

- We additionally need smoothness.

(Bonus) Convergence of Gradient Descent for Strongly Convex Functions

- To establish convergence of m-strongly convex functions:

$$L(v) \geq L(w) + \nabla L(w)^\top (v - w) + \frac{m}{2}\|v - w\|_2^2$$

- We additionally need smoothness. $L(w)$ is M-smooth iff:

$$L(v) \leq L(w) + \nabla L(w)^\top (v - w) + \frac{M}{2}\|v - w\|_2^2$$

(Bonus) Convergence of Gradient Descent for Strongly Convex Functions

- To establish convergence of m-strongly convex functions:

$$L(v) \geq L(w) + \nabla L(w)^\top (v - w) + \frac{m}{2}\|v - w\|_2^2$$

- We additionally need smoothness. $L(w)$ is M-smooth iff:

$$L(v) \leq L(w) + \nabla L(w)^\top (v - w) + \frac{M}{2}\|v - w\|_2^2$$

- Convergence rate:

$$\|w^t - w^*\|_2 \leq \left(1 - \frac{m}{M}\right)^t \|w^0 - w^*\|_2$$

(Bonus) Convergence of Gradient Descent for Strongly Convex Functions

- To establish convergence of m-strongly convex functions:

$$L(v) \geq L(w) + \nabla L(w)^\top (v - w) + \frac{m}{2}\|v - w\|_2^2$$

- We additionally need smoothness. $L(w)$ is M-smooth iff:

$$L(v) \leq L(w) + \nabla L(w)^\top (v - w) + \frac{M}{2}\|v - w\|_2^2$$

- Convergence rate:

$$\|w^t - w^*\|_2 \leq \left(1 - \frac{m}{M}\right)^t \|w^0 - w^*\|_2$$

- Squared loss for linear regression is $\lambda_{\min}$-strongly convex, and $\lambda_{\max}$-smooth

Outline

1. Recap

2. Convexity

3. Closed-form solution for Linear Regression

4. Convergence of Gradient Descent

5. More Losses

More Losses

Function class F
- linear function

Loss function
- different choices

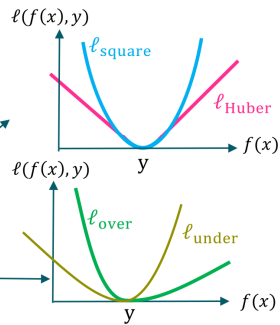
Optimization
- closed-form

So far considered the squared loss

- weighs over- and underestimation the same
- Cost grows quadratically (large errors hugely penalized)

Instead might want:

- ignore outliers (ones with very large penalty) → Huber loss
- weigh over- and underestimation differently → asymmetric losses



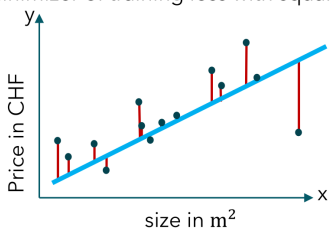
Huber Loss

Function class F
- linear function

Loss function
- Huber loss

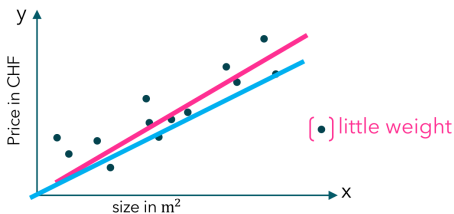
Optimization
- closed-form

Minimizer of training loss with square loss ℓ_{square}



far points → have very little weight

Minimizer of training loss Huber loss ℓ_{Huber}

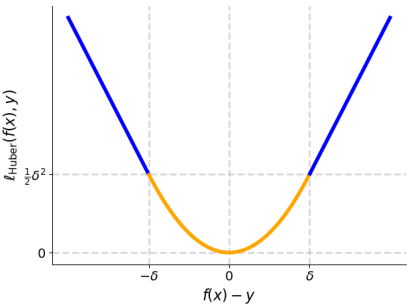


Perhaps better choice if there are outliers in your training data not representing current market

Huber Loss

The Huber loss is given by:

$$\ell_{\text{Huber}}(f(x), y) = \begin{cases} \frac{1}{2}(f(x) - y)^2, & \text{if } |f(x) - y| \leq \delta, \\ \delta|f(x) - y| - \frac{1}{2}\delta^2, & \text{else} \end{cases}$$



See `huber-loss.ipynb`

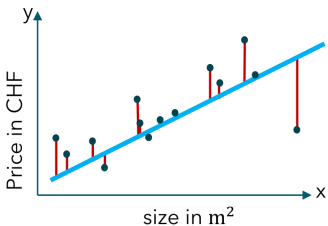
Asymmetric Losses

Function class F
- linear function

Loss function
- asymmetric regression loss

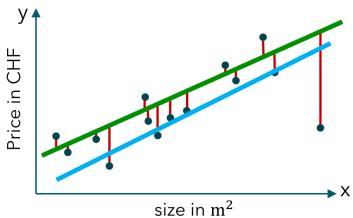
Optimization
- closed-form

Minimum of square loss ℓ_{square}



overestimation penalized less

Minimum of loss that places less weight on overestimation ℓ_{over}

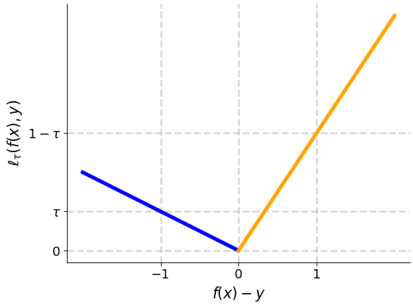


Perhaps better choice if you care more about maximizing profit than how fast you can sell

Asymmetric Losses – Quantile Loss

For $\tau \in (0, 1)$, the Quantile loss is given by:

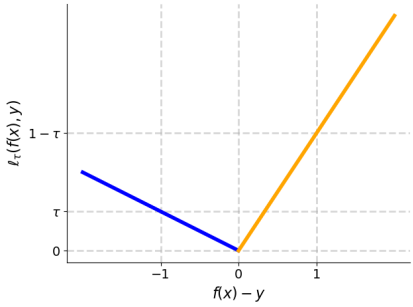
$$\ell_\tau(f(x), y) = \begin{cases} \tau(y - f(x)), & \text{if } f(x) - y < 0, \\ (1 - \tau)(f(x) - y), & \text{else} \end{cases}$$



Asymmetric Losses – Quantile Loss

For $\tau \in (0, 1)$, the Quantile loss is given by:

$$\ell_\tau(f(x), y) = \begin{cases} \tau(y - f(x)), & \text{if } f(x) - y < 0, \\ (1 - \tau)(f(x) - y), & \text{else} \end{cases}$$

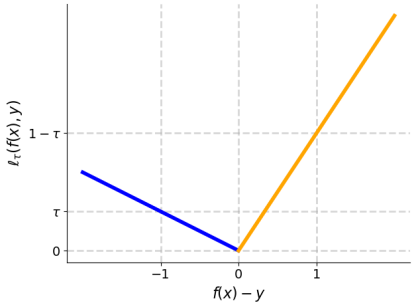


Intuition: small $\tau \to$ underestimation, and large $\tau \to$ overestimation.

Asymmetric Losses – Quantile Loss

For $\tau \in (0, 1)$, the Quantile loss is given by:

$$\ell_\tau(f(x), y) = \begin{cases} \tau(y - f(x)), & \text{if } f(x) - y < 0, \\ (1 - \tau)(f(x) - y), & \text{else} \end{cases}$$



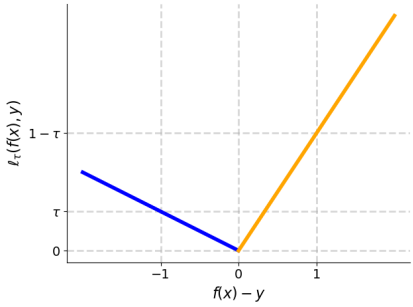
Intuition: small $\tau \to$ underestimation, and large $\tau \to$ overestimation.

(Bonus) For any τ, the minimizer of the quantile loss $f_\tau = \arg\min_f \mathbb{E}[\ell_\tau(f(X), Y)]$:

Asymmetric Losses – Quantile Loss

For $\tau \in (0, 1)$, the Quantile loss is given by:

$$\ell_\tau(f(x), y) = \begin{cases} \tau(y - f(x)), & \text{if } f(x) - y < 0, \\ (1 - \tau)(f(x) - y), & \text{else} \end{cases}$$



Intuition: small $\tau \to$ underestimation, and large $\tau \to$ overestimation.

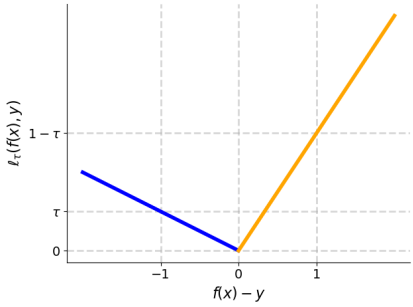
(Bonus) For any τ, the minimizer of the quantile loss $f_\tau = \arg\min_f \mathbb{E}[\ell_\tau(f(X), Y)]$:

- is $f_\tau(x) = q_{P(Y|X=x)}(\tau)$, the conditional quantile function: $P(Y \leq q_P(\tau)) = \tau$

Asymmetric Losses – Quantile Loss

For $\tau \in (0, 1)$, the Quantile loss is given by:

$$\ell_\tau(f(x), y) = \begin{cases} \tau(y - f(x)), & \text{if } f(x) - y < 0, \\ (1 - \tau)(f(x) - y), & \text{else} \end{cases}$$



Intuition: small $\tau \to$ underestimation, and large $\tau \to$ overestimation.

(Bonus) For any τ, the minimizer of the quantile loss $f_\tau = \arg\min_f \mathbb{E}[\ell_\tau(f(X), Y)]$:

- is $f_\tau(x) = q_{P(Y|X=x)}(\tau)$, the conditional quantile function: $P(Y \leq q_P(\tau)) = \tau$
- this implies that f_τ overestimates exactly τ proportion of the points